PROJECT REPORT

FlowMax: Network Flow Optimization System

Project Report submitted for the partial fulfilment of grade for the course

Data Structures and Algorithms - 3 (25CS2103E)

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DECLARATION

The Mini Project Report entitled “FlowMax: Network Flow Optimization System” is a record of Bonafide work of M. Jaswanth – 2520030590, V. Vijay – 2520030387, Sayed Sabeeh – 2520030345 submitted in partial fulfillment for the award of B. Tech in Computer Science and Engineering to the K L University. The results embodied in this report have not been copied from any other departments/University/Institute.

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CERTIFICATE

This is to certify that the mini project based report entitled “FlowMax: Network Flow Optimization System” is a bonafide work done and submitted by M. Jaswanth – 2520030590, V. Vijay – 2520030387, Sayed Sabeeh – 2520030345 in partial fulfillment of the requirements for the award of the degree of BACHELOR OF TECHNOLOGY in Department of Computer Science Engineering, K L (Deemed to be University), during the academic year 2026-2027.

Signature of the Supervisor

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Signature of the External Examiner

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1. ABSTRACT

The FlowMax: Network Flow Optimization System is a Data Structures and Algorithms project designed to determine the maximum possible flow from a source vertex to a destination vertex in a directed network with limited edge capacities. The system uses the Edmonds-Karp algorithm, which is based on the Ford-Fulkerson method and utilizes Breadth-First Search (BFS) to identify augmenting paths efficiently. It maintains a residual network, calculates bottleneck capacities, and updates forward and reverse edge capacities after each iteration while ensuring flow conservation at intermediate vertices.

The system continues this process until no further augmenting path is available, thereby determining the maximum flow. It also identifies the minimum cut and demonstrates the Max-Flow Min-Cut theorem, which establishes the relationship between maximum flow and minimum cut capacity. This project provides a practical understanding of graph representation, queues, residual networks, and network optimization techniques. FlowMax can be applied to real-world problems such as communication bandwidth allocation, traffic management, pipeline distribution, logistics, and resource allocation.

2. INTRODUCTION

The Maximum Flow problem is a fundamental network optimization problem in which the objective is to send the largest possible amount of flow from a source vertex to a destination or sink. Each directed edge has a limited capacity, and intermediate vertices must satisfy flow conservation, meaning the flow entering a vertex equals the flow leaving it.

This project uses the Edmonds-Karp algorithm to repeatedly find augmenting paths and increase the total flow. A residual network is maintained after every augmentation, allowing the system to visualize how routing possibilities change until no further augmenting path exists. The project offers an efficient and systematic approach to solving maximum-flow problems while improving the understanding of graph algorithms and their practical applications.

3. LITERATURE SURVEY

Various approaches have been developed to solve the Maximum Flow problem. The Ford-Fulkerson method uses augmenting paths to increase flow until no suitable path remains. Edmonds-Karp improves this approach by selecting augmenting paths using Breadth-First Search and provides a predictable polynomial-time complexity of $O(VE^2)$.

Residual graphs are essential because they allow previously assigned flow to be reduced or rerouted through reverse edges. The Max-Flow Min-Cut theorem provides an optimality certificate by proving that the maximum flow value is equal to the capacity of a minimum cut.

4. PROBLEM STATEMENT AND OBJECTIVES

The main problem addressed by this project is to develop an efficient network optimization system that determines the maximum flow between a source and destination. The system must respect edge capacities and maintain flow conservation at intermediate vertices.

The primary objectives are:

1. To model a directed capacity network.
2. To implement Edmonds-Karp for maximum-flow computation.
3. To identify augmenting paths and bottleneck capacities.
4. To update the residual network after each augmentation.
5. To calculate the minimum cut.
6. To visualize the flow process.
7. To demonstrate the Max-Flow Min-Cut theorem.

5. HARDWARE AND SOFTWARE REQUIREMENTS

Hardware requirements:

1. Computer or laptop with an Intel Core i3 processor or above.
2. Minimum of 4 GB RAM.
3. At least 500 MB of available storage space.
4. Keyboard, mouse, and monitor or display.

Software requirements:

1. Operating system: Windows or Linux.
2. Programming language: Java (JDK 8 or above).
3. IDE or code editor: Eclipse, IntelliJ IDEA, or Visual Studio Code.
4. Standard Java data structures such as arrays, lists, queues, and hash maps for graph and residual-network representation.
5. Java Swing or JavaFX for interactive network visualization.

6. RESULTS

The developed FlowMax system successfully calculates the maximum flow for the given network using Edmonds-Karp. The system identifies each augmenting path, calculates its bottleneck capacity, updates the forward and reverse residual capacities, and continues until no source-to-sink augmenting path remains.

The final output includes the maximum flow value, flow assigned to network edges, the final residual network, and the minimum-cut capacity. The visualization makes the optimization process easier to understand and provides a direct verification of the Max-Flow Min-Cut theorem.

7. CONCLUSION

The FlowMax project successfully implements the Maximum Flow problem using the Edmonds-Karp algorithm. The proposed approach models capacity-constrained directed networks, finds augmenting paths, maintains flow conservation, and updates residual capacities after each iteration. It determines the maximum transferable flow and identifies a corresponding minimum cut.

The project demonstrates the practical application of graph algorithms, queues, residual networks, optimization, and the Max-Flow Min-Cut theorem. Overall, the system provides both an efficient solution and a clear visual explanation of the maximum-flow process.

8. FUTURE SCOPE

The future scope of this project includes extending FlowMax to handle larger and more complex networks using optimized graph representations. The system can be enhanced with an interactive JavaFX or web-based interface for adding nodes, editing capacities, and selecting source and sink vertices.

Animated augmenting paths and residual-network transitions can provide better visualization of the algorithm. Future versions can compare Ford-Fulkerson and Edmonds-Karp with other network optimization techniques and support real-world applications such as communication bandwidth allocation, traffic routing, pipeline distribution, logistics, and resource allocation.